PINNStudio, tuned (forward, inverse) | Starting settings: DeepXDE's example / the paper's method | Both the same (no tuning needed) | Reported in the paper

**ODEs**
- Integro-differential ODE
- Integro-differential ODE – Volterra kernel
- Linear ODE system — $\omega$
- Lotka-Volterra ODE — $d$, $a$
- Second-order ODE — $a$, $b$
- Lorenz ODE – unknown $C_1$, $C_2$, $C_3$ — $C_1$, $C_3$, $C_2$
- Lorenz ODE – unknown $C_1$, $C_2$, $C_3$, exogenous input — $C_3$, $C_2$, $C_1$

**1D, steady**
- Euler beam 1D — $EI$
- Fractional Poisson 1D
- Poisson 1D – Dirichlet
- Poisson 1D – Dirichlet/Neumann
- Poisson 1D – Dirichlet/Periodic
- Poisson 1D – Dirichlet/point-set operator
- Poisson 1D – Dirichlet/Robin
- Poisson 1D – hard BCs
- Poisson 1D – multi-scale Fourier features
- Brinkman-Forchheimer 1D – unknown v_e, K — $K$, $\nu_e$
- Diffusion-reaction 1D – unknown rate field — $k(x)$
- Fractional Poisson 1D – unknown order α — $\alpha$
- Poisson 1D – unknown forcing field — $q(x)$
- Poisson 1D – unknown forcing field (batches) — $q(x)$

**1D, time-dependent**
- Allen-Cahn 1D – Wight & Zhao — $\gamma_2$, $\gamma_1$
- Cahn-Hilliard 1D – Wight & Zhao
- Cahn-Hilliard 1D – Mattey & Ghosh
- Black-Scholes 1D – American put
- Black-Scholes 1D – European call
- Burgers 1D – Raissi et al. — $\lambda_1$, $\lambda_2$
- Burgers 1D – RAR
- Diffusion 1D
- Diffusion 1D – hard IC and BCs
- Diffusion 1D – resampling
- Diffusion-reaction 1D — $D$
- Fractional diffusion 1D
- Heat 1D – Dirichlet — $a$
- Heat 1D – resampling
- Klein-Gordon 1D — $\gamma$
- Schrödinger 1D — $c$
- Wave 1D – space-time multi-scale Fourier features — $C$
- Diffusion 1D – unknown coefficient — $C$
- Diffusion-reaction 1D – unknown k_f, D — $D$, $k_f$

**2D, steady**
- Fractional Poisson 2D – disk
- Helmholtz 2D – Dirichlet — $k_0$
- Helmholtz 2D – hyperparameter search
- Helmholtz 2D – sound-hard scattering
- Helmholtz 2D – square with hole
- Laplace 2D – disk
- Linear elasticity 2D – plate — $\mu$, $\lambda$
- Linear elasticity 2D – plate, soft BCs
- Navier-Stokes 2D – Kovasznay flow — $Re$
- Poisson 2D – disk — $f$
- Poisson 2D – L-shape
- Fractional Poisson 2D – unknown order α — $\alpha$

**2D, time-dependent**
- Allen-Cahn 2D – Wight & Zhao — $\lambda\varepsilon^2$, $\lambda$
- Allen-Cahn 2D – Mattey & Ghosh — $c_2$, $c_1^2$
- Cahn-Hilliard 2D – Wight & Zhao — $\varepsilon^2$
- Cahn-Hilliard 2D – Mattey & Ghosh — kappa, $\alpha$
- Burgers 2D – Mathias et al. — $\nu$
- Heat 2D — $\alpha$

**3D**
- Fractional Poisson 3D – sphere
- Poisson 3D – sphere — $f$
- Allen-Cahn 3D – Wight & Zhao — $\lambda\varepsilon^2$, $\lambda$
- Heat 3D — $\alpha$
- Navier-Stokes 3D – Beltrami flow — $\nu$

$10^{-6}$ $10^{-5}$ $10^{-4}$ $10^{-3}$ $10^{-2}$ $10^{-1}$ $10^{0}$

Relative $L_2$ error of the main output at the last reference time

0.001 0.1 10

Parameter (or unknown field) error (%)